Regional and Trade Dimensions of Climate Change

Authored by Professor Bartholomew Armah (PhD)
Senior Fellow
African Center for Economic Transformation (ACET)

Introduction

Climate change poses challenges for Africa's development but it also offers opportunities particularly through international trade which accounted for 22 percent (i.e., $689 billion in merchandise exports and imports) of the continent's GDP of $3.1 trillion in 2023.[1] On the one hand, the destructive impacts of climate change on infrastructure, agricultural yields and GDP can derail Africa's sustainable development aspirations. On the other hand, by enhancing access to low-carbon technologies and facilitating Africa's participation in green value chains, international trade offers the continent opportunities for decarbonization and green growth. Trade in environmental goods and services also offers an opportunity for Africa to leverage its critical minerals to finance its development objectives.

The transition to a low-carbon economy will, however, involve trade-offs particularly for developing countries. Without international support, the initial costs associated with investments in environmental technologies, products and services and the potential loss of competitiveness and revenue associated with the imposition of taxes on the continent's carbon-intensive products will, in the short-run, undermine incomes and welfare in developing economies. For instance, the African Continental Free Trade Area is expected to increase the volume of Africa's GDP by 0.9 percent. However, imposing a $25 tax on carbon is estimated to reduce this increase to 0.3 percent (ECA et al 2024).

Hence, managing the trade-offs associated with the low-carbon transition and leveraging the synergies between decarbonization and trade are key to a sustainable decarbonization agenda. Appropriately designed trade policies and carbon pricing regimes, complemented by climate financing at scale can expedite decarbonization by facilitating the production and exchange of environmentally friendly goods, services and technologies, and dis-incentivizing the production of carbon-intensive products.

Climate change and international trade

Climate change disrupts international trade. Climate-induced floods and drought damage transport, agricultural produce, energy infrastructure, and natural habitats and ecosystems resulting in disruptions of supply chains and productivity losses. A 1°C temperature increase is estimated to reduce export volumes by between 5 and 6 percent (WTO, 2022).

On the other hand, international trade can strengthen climate resilience and mitigation by facilitating access to environmental technologies; lowering the cost of renewable energy products through increased market access; improving standards and policies that promote trade in energy efficient systems; encouraging country specialization based on capabilities and efficiencies; and nurturing global supply

[1] In 2023, merchandise exports ($579.167bn) and imports ($109.856bn), accounted for 22 per cent of Africa's GDP ($3.1 trillion) in current US dollars. https://unctadstat.unctad.org/datacentre/dataviewer/US.IntraTrade. Accessed 12/9/2024.

chains that encourage sustainable production practices. Improving the sustainability of supply chains could reduce CO2 emissions by approximately 2 billion tons annually by 2030 (Carbon Trust) and international trade could reduce emissions by up to 1.4 gigatons of CO2 annually if countries specialize according to their capabilities and efficiencies (WTO, 2022).

Harnessing the positive effects of trade through appropriate pricing

Harnessing the positive effects of trade on climate change requires policy actions that ensure appropriate pricing of carbon-intensive and low-carbon products and create an enabling regulatory environment for the production and exchange of environmentally friendly technologies, services and goods (Stern 2006, p. xxv). A removal of tariffs and Non-Tariff Barriers for four basic clean energy technologies (wind, solar, clean coal, and efficient lighting) in 18 of the high-GHG-emitting developing countries will result in trade gains of up to 13 percent (World Bank 2008).

Pricing carbon-intensive goods and services

Furthermore, fossil fuel-based products enjoy a price advantage over environmentally friendly products because they do not price in the negative climate effects associated with their production and consumption. Furthermore, carbon-intensive products often benefit from universal subsidies which fuel their consumption even as environmental technologies and products are subjected to trade barriers.

Establishing a uniform global price (per ton) of carbon emissions and imposing a tax on carbon-intensive products based on the established uniform price would enhance the relative competitiveness of environmental goods and services (EGS), reduce the likelihood of carbon leakage and minimize potential violations of WTO protocols on the grounds of discriminatory trade practices.

However, there is currently no global agreement on carbon pricing; up to 70 separate pricing initiatives have been adopted in 46 national jurisdictions worldwide (WTO, 2022) [2]. Furthermore, existing carbon pricing schemes cover only 23 per cent of total carbon emissions and less than 4 per cent of global emissions is currently covered by a carbon price in the range needed by 2030, to prevent the average global temperature from increasing by 2°C (World Bank, 2022). Artificially low carbon taxes are unlikely to curb the fossil-fuel appetite of firms and consumers.

However, low-income countries which tend to be net-exporters of carbon-intensive products, could be adversely affected by a tax on carbon which is essentially a tariff on the exports of such products.[3] These adverse effects could nevertheless be mitigated by scaling-up the design of credible carbon offset projects in developing countries to benefit from established carbon prices. The caveat is that carbon credits must be competitively priced and communities must be equitably compensated (Böhm and Dabhi (eds) 2009). Currently, carbon credits generated by carbon offset projects attract much lower carbon prices than developed countries. In developing countries, the average price for carbon credits, particularly those generated through the Clean Development Mechanism (CDM), has historically ranged from $2 to $10 per ton of CO2 equivalent (tCO2e). This is much lower than the price of US$50–100/t$CO_2$e proposed by the

[2] The IMF estimates that an average price of $75 per ton of CO2 is needed to limit global warming to 2°C above pre-industrial levels. The Report of the High-Level Commission on Carbon Prices (2017) estimated that the appropriate global carbon price should be US$40–80/t$CO_2$e by 2020, and US$50–100/tCO_2e by 2030.

[3] The adverse effects of carbon pricing are not specific to Africa, even a globally coordinated uniform carbon price is estimated to reduce global GDP by 0.46 percent rising to 0.68 percent in an uncoordinated carbon pricing regime.

High-Level Commission on Carbon to limit global warming to 2°C above pre-industrial levels (Report of the High-Level Commission on Carbon Prices, 2017).

Implementing a uniform carbon price would eliminate the variance in carbon prices and increase the price per ton of carbon (or value of carbon credits) received for carbon offset projects in developing countries. Redirecting the revenues generated by carbon tax initiatives, such as the European Union's Carbon Border Adjustment Mechanism (CBAM), to a fund dedicated to financing the low-carbon transition of developing countries would also be helpful in smoothing the low-carbon transition of such countries.

Incentivizing the production of low-carbon goods and services through regulatory instruments

Beyond fiscal measures (i.e., tariffs and taxes), regulatory instruments such as regulations, directives, standards and mandates can be used to incentivize the production and trade of environmentally friendly technologies, products and services. This is achieved through strict efficiency requirements, ambitious renewable energy targets requiring firms to source a minimum percentage of their energy from renewable sources. In January 2024 Ethiopia became the first country to ban the import of internal combustion engine vehicles effectively forcing a shift toward electric vehicles (International Trade Administration, 2024). Regulatory instruments have implications for the composition of a country's trade basket. Manufacturers are forced to import electric vehicles and renewable energy related technologies and equipment to comply with the regulatory instruments with adverse implications in the short term. Indeed, some studies find that both carbon taxes and energy efficiency standards have a statistically significant negative effect on competitiveness through their impact on bilateral trade. In practice, however, countries offset the potentially negative impacts of regulatory instruments with generous policy packages, including exemptions, which ultimately negate their effectiveness in encouraging low-carbon production pathways (World Bank 2008).

Leveraging the African Continental Free Trade Area for decarbonization

The African Continental Free Trade Area is the largest free trade area in the world, comprising a population of 1.4bn people and a $2.4 trillion market. The AfCFTA is estimated to increase the continent's GDP by 0.9 percent (i.e., US$113 billion) by 2045. The Agreement includes provisions to address climate change. Article 26 of the AfCFTA investment protocol enjoins member States to promote investments in climate mitigation and adaptation programmes and to mitigate greenhouse gas emissions and climate change impacts on exhaustible natural resources such as fresh water and biological diversity.

By reducing trade barriers, enhancing access to a huge ($2.4 trillion) market and fostering trade-facilitating infrastructure development, the AfCFTA can attract and scale-up up green investments leading to more competitive prices, expedited transfer of green technologies and much needed employment growth. However, under-investments in trade-facilitating infrastructure - particularly roads, railways and airports - and poor logistics, and the absence of harmonized cross border standards have contributed to high transport costs, slow clearance of cargo at ports and inadequate interconnectivity with other infrastructure networks on the continent (Africa Finance Cooperation, 2024). These challenges undermine the continent's capacity to realize the full potential of the AfCTA for a low-carbon transition.

The AfCFTA also highlights the short-term trade-offs associated with low carbon transitions. There is evidence to suggest that while implementation of Nationally Determined Contributions and carbon tax

policies by AfCFTA signatories reduces GHG emissions by 24.8 percent, it also reduces potential export growth from 5.8 percent to 3.5 percent (ECA, ATPC, CEPII and DANIDA (2024). This finding again reinforces the need for international measures to support the continent's low-carbon transition.

Conclusion: Minimizing decarbonization trade-offs for a truly just transition

Enhancing access by developing countries to climate friendly technologies, scaling up concessionary climate financing and strengthening capacities to adapt and mitigate climate change are critical to minimizing decarbonization trade-offs and securing a truly just transition. Measures to scale up concessionary financing include de-risking private sector investments in green technologies, encouraging debt for climate adaptation swaps to address debt sustainability and redirect resources into climate finance, leveraging carbon offset projects to mobilize climate finance and securing the commitment of advanced countries to a clear roadmap to deliver a minimum of US$ 100bn in annual climate financing. International cooperation is also needed to scale climate finance, facilitate technology transfer and achieve uniform pricing of carbon.

To mitigate the adverse impacts of carbon taxes on developing countries, a portion of revenues accruing from carbon border adjustment initiatives could be redirected to a climate fund to smooth the low-carbon transition of developing countries. North-South cooperation complemented by South-South cooperation offers unique opportunities for developing countries to share technologies and enhance access to development financing since approximately 70 percent of SSC initiatives already involve training, project financing and technology transfer.

International trade can play a pivotal role in ensuring a successful decarbonization agenda. However, managing the trade-offs associated with the low-carbon transition and leveraging the synergies between decarbonization and trade are key to a sustainable decarbonization agenda.

References

1. Africa Finance Cooperation (2004) State of Africa' Infrastructure Report 2024. The Infrastructure Imperative: Igniting Africa's Industrial Renaissance.
2. Steffen Böhm and Siddhartha Dabhi (eds) (2009), Upsetting the Offset: The Political Economy of Carbon Markets. First published by MayFlyBooks in paperback in London and free online at www.mayflybooks.org in 2009. Printed by the MPG Books Group in the UK
3. ECA, ATPC, CEPII and DANIDA (2024). Greening the implementation of the Africa Free Trade Agreement
4. International Trade Administration (2024), Market Intelligence: Ethiopia Automotive EV Market. https://www.trade.gov/market-intelligence/ethiopia-automotive-ev-market. Accessed December 13 2024.
5. Stern, N. 2006. Stern Review on the Economics of Climate Change. London: HM Treasury, U.K.
6. The World Bank (2008). International Trade and Climate Change: Economic, Legal and Institutional Perspectives. Washington, DC.
7. World Trade Organization (2022) World Trade Report 2022: Climate Change and International Trade.

Recommended Reading

1. The European Carbon Market in Action: Lessons from the First Trading Period" by Ellerman, Convery, and De Perthuis (2010): This book provides an in-depth analysis of the European Union Emissions Trading Scheme (EU ETS), discussing how the market has evolved and affected carbon prices.
2. Carbon Markets: An International Business Guide" by Arnaud Brohé, Nick Eyre, and Nicholas Howarth (2009): This source provides insights into how carbon markets function internationally, including pricing and regulatory frameworks.
3. Economics of Carbon Markets: Taming the Triple Challenge" by Axel Michaelowa (2020): The paper explores economic principles behind carbon pricing and market mechanics, focusing on the effectiveness of different pricing mechanisms.
4. "Carbon markets in a climate-changing capitalism" by Gareth Bryant (2019): This study critically examines how carbon markets operate within the broader economic system, with a focus on pricing dynamics and market design.
5. "The impact of the EU ETS on firm profits and market prices" by Hintermann, Jaraite, & Karlsson (2016): This study assesses how carbon pricing under the EU ETS has influenced firm behavior and market outcomes, including price formation.

Definitions

1. **Carbon tax.** A carbon tax is a tax on the carbon content of fuels (principally coal, oil, and natural gas) that generate CO2 emissions when burned. The tax would apply at a specific rate per ton of coal, per barrel of oil, or per million cubic feet of gas, with the amounts adjusted to equalize implied taxes on carbon content.
2. **Market-Based Instruments**
   These instruments are based on the premise that "free markets find optimal solutions." They include emissions trading and tradable renewable energy certificates (TRCs) as effective means to help decrease the cost of mitigating greenhouse gas emissions.

3. **Voluntary Agreements (VAs)**
   Voluntary agreements differ from other measures in that they are negotiated directly between governments and industry/firms rather than result from mandates imposed by the governments; they are often the preferred policy approach from industries' perspective. (World Bank 2008, Appendix
4. **Carbon leakage**
   Carbon leakage refers to the situation that may occur if, for reasons of costs related to climate policies, businesses were to transfer production to other countries with laxer emission constraints. This could lead to an increase in their total emissions. The risk of carbon leakage may be higher in certain energy-intensive industries.
5. **Carbon Offsets**
   Carbon offsets are projects that reduce, avoid, or sequester an equivalent amount of carbon dioxide (CO2) or other greenhouse gases (GHGs) elsewhere. Carbon offsets generate carbon credits that represent the reduction or removal of one metric ton of carbon dioxide (CO2) or its equivalent in other greenhouse gases.
6. **Carbon Border Adjustment .**
   Such mechanisms are intended to impose tariffs on imported goods based on the difference between the producer's level of GHG pricing and the carbon price in the importing jurisdiction.
7. **Greenhouse Gases**
   GHGs comprise carbon dioxide (CO2), methane (CH4), nitrous oxide (N2O) and fluorinated gases, including hydrofluorocarbons (HFCs), perfluorocarbons (PFCs) and sulphurhexafluoride (SF6). Although discussions on climate change tend to focus on CO2 because it is the main contributor to climate change, accounting for about three-quarters (74.1 per cent) of total emissions, it is estimated that methane contributes 17.3 per cent, nitrous oxide 6.2 per cent, and other emissions 2.4 per cent (WRI, 2022).
8. **Environmental Goods and Services (EGS)**
   EGS are defined as goods and services used to measure, prevent, limit, minimize or correct environmental damage to water, air and soil, as well as problems related to waste, noise and eco-systems (OECD and Eurostat, 1999). They include cleaner technologies, products and services that reduce environmental risks and minimize pollution and resource use.